A partial harmonic sum is never an integer

Ireland–Rosen, 1.30

Prove that, for $n \geq 2$,

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

is not an integer.

Proof. Suppose, to the contrary, that for some integer m,

$$\sum_{k=2}^{n} \frac{1}{k} = m$$

as rational numbers.

By Bertrand's postulate applied to $n/2$, which is positive since $n \geq 2$, there exists a prime p such that

$$\frac{n}{2} < p \leq 2 \cdot \frac{n}{2}.$$

Hence $p \leq n$, $1 < p$, and $n < 2p$. In particular, p occurs among the integers $2, \ldots, n$.

We first observe that no other integer k with $2 \leq k \leq n$ is divisible by p. Indeed, if $p \mid k$, write $k = pc$. Since $k > 0$, we have $c > 0$. If $c \geq 2$, then $k = pc \geq 2p > n$, contradicting $k \leq n$. Thus $c < 2$, so $c = 1$ and $k = p$.

More generally, if s is a finite set of natural numbers none of which is divisible by p, then p does not divide $\prod_{k \in s} k$. This follows by induction on s: the empty product is 1, which is not divisible by p since $p > 1$; and if the product over $s \cup \{a\}$ is $a \prod_{k \in s} k$, then primality of p implies that p dividing this product would force p to divide a or $\prod_{k \in s} k$.

Now put

$$M = \prod_{\substack{2 \leq k \leq n \\ k \neq p}} k.$$

By the preceding observations, $p \nmid M$. Also, each k in this product divides M, so M/k is a natural number for $2 \leq k \leq n$, $k \neq p$. Define the natural number

$$S = \sum_{\substack{2 \leq k \leq n \\ k \neq p}} \frac{M}{k}.$$

Since $k \mid M$, we have, as rational numbers,

$$(S : \mathbb{Q}) = \sum_{\substack{2 \leq k \leq n \\ k \neq p}} \frac{1}{k} \cdot M.$$

Multiplying our assumed equality by M and separating the term $k = p$ gives

$$\begin{aligned}\frac{1}{p} \cdot M + S &= \frac{1}{p} \cdot M + \sum_{\substack{2 \le k \le n \\ k \neq p}} \frac{1}{k} \cdot M \\ &= \sum_{k=2}^{n} \left(\frac{1}{k} \cdot M \right) \\ &= \left(\sum_{k=2}^{n} \frac{1}{k} \right) M \\ &= mM.\end{aligned}$$

Thus, with the integer

$$z = mM - S,$$

we obtain

$$\frac{1}{p} \cdot M = z$$

as rational numbers. Multiplying by p yields $M = zp$ as rational numbers, hence also as integers:

$$(M : \mathbb{Z}) = z \cdot (p : \mathbb{Z}).$$

Therefore $(p : \mathbb{Z}) \mid (M : \mathbb{Z})$, and consequently $p \mid M$ as natural numbers. This contradicts $p \nmid M$.

Hence $\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ is not an integer. □